Day 3: Reinforcement Learning with Large Language Models

Study notes • Language models, sampling, RL mathematics, PPO, GRPO, GSPO/SAPO overview, and vLLM

Note: These are newly prepared study notes aligned with your day-wise schedule, not pages extracted from the original AMD PDF.

1. Learning objectives

• Explain how reinforcement learning can fine-tune a language model toward preferred responses.

• Understand policy, action, trajectory, reward, return, advantage, and reference policy.

• Describe PPO and GRPO at a conceptual level and identify the role of inference engines such as vLLM.

2. Language-model preliminaries

• An autoregressive language model predicts a probability distribution for the next token, then repeats the process for each generated token.

• Logits are raw model scores; softmax converts them into probabilities. The prompt, model parameters, and decoding settings affect the output.

• Greedy decoding chooses the highest-probability token. Temperature changes the sharpness of the distribution. Top-k restricts candidates to k tokens; top-p keeps candidates until cumulative probability reaches a threshold.

• Sampling matters in RL because the system may generate multiple candidate responses and compare their rewards. Different sampling settings can change diversity and reward estimates.

3. Core reinforcement-learning terms

• Term | Meaning

• Policy | The model that chooses tokens or responses.

• State / context | The prompt and generated tokens so far.

• Action | A selected token or decision.

• Trajectory | A complete generated response or action sequence.

• Reward | A score reflecting task success or preference.

• Return | Reward accumulated over a trajectory.

• Advantage | How much better an action is than a baseline expectation.

• Reference policy | A fixed model used to constrain policy drift.

4. Reward models and preference data

• Reward models estimate how desirable a response is, often from human preference comparisons between responses.

• Preference datasets commonly pair a prompt with two candidate answers and indicate which is preferred.

• Rewards are imperfect proxies for quality. A model may exploit weaknesses in the reward model, so independent evaluations remain important.

• Rewards may assess the final outcome or intermediate steps; the choice depends on the task and training design.

5. PPO: Proximal Policy Optimization

• PPO is a policy-gradient algorithm used in some LLM alignment pipelines. It combines sampled responses, reward estimates, advantage estimates, and a clipped policy objective.

• Typical workflow: generate responses; score them; estimate advantages; update the policy while limiting overly large changes; evaluate the result.

• Clipping discourages a single update from changing action probabilities too sharply. It can improve stability but does not guarantee training will always be stable.

• A reference-policy penalty or related constraint may help prevent the tuned model from drifting too far from the original model.

6. GRPO, GSPO and SAPO

• Group Relative Policy Optimization (GRPO) samples several responses for the same prompt and uses their relative rewards to estimate which responses are better.

• A simplified GRPO workflow is: sample a group; score each response; calculate relative or normalized advantages; update the policy under appropriate constraints.

• Group-relative baselines can reduce reliance on a separately learned value model in some implementations.

• GSPO and SAPO are additional optimization approaches named in the schedule. Their exact equations and behavior depend on the specific paper and implementation; do not assume they are identical to PPO or GRPO.

7. vLLM and efficient inference

• vLLM is an inference and serving engine intended to improve throughput and GPU memory management for supported language models. Its PagedAttention approach manages key-value cache memory in blocks.

• RL training can require generating many candidate responses, making inference a major bottleneck. Faster inference can shorten the data-generation stage.

• vLLM is an inference engine, not an RL algorithm. Performance depends on model, hardware, batch size, sequence length, precision, and configuration.

8. Evaluation and failure modes

• Track task accuracy and independent evaluations as well as reward scores; reward improvement alone may be misleading.

• Monitor response length, diversity, refusal behavior, and safety regressions.

• Common risks include reward hacking, mode collapse, unstable updates, over-optimization, and data leakage.

• Use held-out prompts and compare the tuned model with base and supervised fine-tuned baselines.

Practical Exercises

1. Write pseudocode for sampling four answers to a prompt, scoring them, and comparing rewards.

2. Compare temperature, top-k, and top-p on the same prompt; record differences in variety.

3. Create an evaluation checklist covering accuracy, reward, response length, and safety.

Review Questions

1. What is the difference between a policy and a reward model?

2. Why might maximizing reward fail to improve real task quality?

3. Describe the main stages in a PPO-style LLM training loop.

4. How does GRPO use a group of responses?

5. What problem does vLLM help solve?

6. Why are independent evaluations needed alongside reward scores?